

		momentum $p=mv = 1.8 \times 5.1 = 9.18 \text{ kg m s}^{-1}$
6.	B	At maximum height, initial horizontal velocity = $50 \cos 80^\circ = 8.62 \text{ m s}^{-1}$ At this instance when the missile explodes. Total vertical momentum is zero By conservation of momentum, total initial momentum (hor) = total final momentum of fragment Z (hor) $3M (8.62) = Mv \text{ (of fragment z)} \quad v = 25.9 \text{ m s}^{-1}$
7.	D	Resolving forces vertically